

How a school swarms

collective motion for the aquarium level's ten followers · the Couzin zone model, measured against our own Forth implementation

Lengths are in **body lengths** (BL; a fish is 3.5 in). The tank is $13.4 \times 3.4 \times 4.7$ BL inside. A **school** is polarised (p_group near 1); a **swarm** is not (p near 0.1): moving one zone width, Δr_o , switches between them.

A · Four collective states from one rule, simulated here (N = 100, the paper's r_r , α , θ , s , σ)

A · swarm

top view

$\Delta r_o = 0, \Delta r_a = 15$
p_group 0.16 · m_group 0.10

ours · N = 100 simulated here; the zone widths are ours, the paper prints none

B · torus

spin-axis view

$\Delta r_o = 1, \Delta r_a = 12$
p_group 0.09 · m_group 0.90

C · dynamic parallel

side view

$\Delta r_o = 3, \Delta r_a = 10$
p_group 0.90 · m_group 0.03

D · highly parallel

side view

$\Delta r_o = 10, \Delta r_a = 10$
p_group 1.00 · m_group 0.02

B · The model: three zones and a blind volume

repulsion zone, $r_r = 1$
anyone here: turn away, nothing else counts

orientation zone, width Δr_o
those here: match their heading

attraction zone, width Δr_a
those here: turn toward them

blind volume, 90° behind
 $\alpha = 270^\circ$: cannot see here

Drawn at the school setting: $\Delta r_o = 5, \Delta r_a = 6$

verified · zones, α , r_r : Couzin 2002

ours · widths drawn

C · The map: p_group over the two zone widths

p_group (alignment)

< 0.25 < 0.6 < 0.9 ≥ 0.9

S swarm: low p, low m
T torus: low p, high m
D dynamic parallel: high p
P highly parallel: p ≈ 1
F fragments (more than half)
? seeds disagree: no one state

our swarm, torus, school

ours · N = 100, 3 replicates, 700 steps (paper: 30)

D · The rule in eight lines, and the Forth word that does each

1 See the others, except behind

2 Repulsion if anyone is within r_r

3 Else orientation and attraction

4 Nobody in any zone: keep the heading

5 Turn toward d by at most $\theta\tau$, move s τ

6 The leader counts w times in sums 3

7 Walls repel like a neighbour

8 A dart startles the followers nearby

$u_{ij} \cdot v_i \geq \cos(\alpha/2)$

$d = -\sum u_{ij}$ (nothing else counts)

$d_o = \sum v_j, d_a = \sum u_{ij}; d = (d_o + d_a)/2$

$d = v_i$

$v \leftarrow \text{rotate}(v, d, \theta\tau); c \leftarrow c + s \tau v$

$\sum w_j v_j, \sum w_j u_{ij}$

wall within 0.6 BL: one more repulsion

swim away from the leader for 0.5 s

sch-pair

sch-pair

sch-pair, sch-want

sch-want

sch-turn, sch-follow

sch-pair (weight)

sch-wall

sch-startle-all

The paper's random error ($\sigma = 0.05$ rad) is left out of the Forth: the caller adds it. Our units differ from the paper's, on purpose: at its $s = 3$ BL/s and $\theta = 40^\circ/s$ a 90° turn takes 6.75 BL of swimming (radius 4.3 BL), wider than the tank is deep (3.4 BL), and the tank runs below could not keep the fish in. Ours: $s = 2$ BL/s, $\theta = 120^\circ/s$, 1.50 BL, radius 0.95 BL.

verified · rules 1–5 and the paper's s, θ : Couzin 2002, eqns 1–3 and Fig. 3

ours · rules 6–8, our s, θ , the 0.6 BL wall, the 0.5 s startle

F · The Forth core, school.fth: the four rule words in full (the rest is in the file), and what it costs

```
: sch-pair
dup 19 sc!
0 over sch@ 0 cur sch@ - 16 sc! 1 over sch@ 1
cur sch@ - 17 sc! 2 swap sch@ 2 cur sch@ -
18 sc!
16 vlen2 dup 0.000001 > over 1 par@ 0 par@ + 2
par@ + dup * < & if
sch-sqrt dup 23 sc! 1 swap / 16 vscale
heading-dot-u 3 par@ >= if
23 sc@ 0 par@ < if
-1 0 acc+u 1 9 sc+!
else 23 sc@ 0 par@ 1 par@ + < if
19 sc@ weight 3 acc+v 1 10 sc+!
else
19 sc@ weight 6 acc+u 1 11 sc+!
then then
then
else drop then ;
: sch-want
12 vzero
9 sc@ 0 > if 0 vnorm drop 0 sc@ 12 sc! 1 sc@ 13
sc! 2 sc@ 14 sc!
else
10 sc@ 0 > if 3 vnorm drop 3 sc@ 12 sc+! 4 sc@
13 sc+! 5 sc@ 14 sc+! then
11 sc@ 0 > if 6 vnorm drop 6 sc@ 12 sc+! 7 sc@
13 sc+! 8 sc@ 14 sc+! then
then
12 vnorm 0.000001 > not if 3 cur sch@ 12 sc! 4 cur
sch@ 13 sc! 5 cur sch@ 14 sc! then ;
: sch-turn
3 cur sch@ 12 sc@ * 4 cur sch@ 13 sc@ * + 5 cur
sch@ 14 sc@ * + dup 4 par@ >= if
drop
else
>r
12 sc@ r@ 3 cur sch@ * - 20 sc! 13 sc@ r@ 4
cur sch@ * - 21 sc! 14 sc@ r> 5 cur sch@ *
- 22 sc!
20 vnorm 0.000001 > not if 4 cur sch@ 20 sc! 3
cur sch@ negate 21 sc! 0 22 sc! 20 vnorm
0.000001 > not if 1 20 sc! 0 21 sc! then
then
3 cur sch@ 4 par@ * 20 sc@ 5 par@ * + 12 sc!
4 cur sch@ 4 par@ * 21 sc@ 5 par@ * + 13 sc!
5 cur sch@ 4 par@ * 22 sc@ 5 par@ * + 14 sc!
12 vnorm drop
then ;
: sch-follow
15 sc! 0 vzero 3 vzero 6 vzero 0 9 sc! 0 10 sc! 0
11 sc!
sch-n 0 do i cur <> if i sch-pair then loop
sch-walls sch-want
sch-startle? if sch-startle-away sch-startle-tick
then
sch-turn
12 sc@ 9 cur sch! 13 sc@ 10 cur sch! 14 sc@ 11
cur sch!
0 cur sch@ 9 cur sch@ 6 par@ * + 6 cur sch!
1 cur sch@ 10 cur sch@ 6 par@ * + 7 cur sch!
2 cur sch@ 11 cur sch@ 6 par@ * + 8 cur sch! ;
```

Dictionary bytes per word (zForth HERE), 2179 B in all

sch-turn 283

sch-pair 240

sch-follow 211

sch-want 201

sch-startle-all 123

28 other words 1121

2179 B

6.2 ms

1e-03

~0.6 ms

of the 64 KB dictionary (3.3 %); 79 code lines

one tick, 11 fish, on the Chromecast HD (median of 3 runs of 200); x86 here: 7.0 ms

largest heading error against couzin.py in one tick (the engine's sine is 0.2 % off)

per follower per tick: spread over frames, the full 10 Hz step is about 6 % of a core

ours · the Forth, its size, error and timing are our measurements: the interpreter only, not yet inside the engine

G · The Forth, running in the tank: swarm, torus, school, and a startle

swarm setting (0, 10)

torus setting (1, 10)

school setting (5, 6)

Startle: distance to the leader

leader weight w

dist · align · p · in box

w = 1 2.7 BL · +0.02 · 0.27 · yes

w = 3 2.8 BL · -0.12 · 0.40 · no

w = 6 2.5 BL · +0.07 · 0.51 · no

w = 1 3.4 BL · -0.05 · 0.58 · yes

w = 3 2.3 BL · +0.24 · 0.64 · no

w = 6 2.3 BL · +0.31 · 0.61 · no

w = 1 3.8 BL · +0.12 · 0.87 · yes

w = 3 3.5 BL · +0.17 · 0.84 · yes

w = 6 3.2 BL · +0.18 · 0.82 · yes

ours · the Forth core in the tank's real box (w = 3, seed 1 drawn; the table is the mean of 3 seeds; "no" = a fish strayed more than 0.5 BL out)

Sources · live links in the PDF; what could not be verified is marked

S1 Couzin et al. 2002, J. Theor. Biol. 218:1–11

verified

zone model, four states, p_group, m_group, Fig. 3 (PDF read)

S2 Couzin et al. 2005, Nature 433:513–516

unverified

informed minority guides a group (summary only)

S3 Pitcher 1983, via Wikipedia

unverified

shoal to school is a continuum (secondary sources)

S4 Ocellaris clownfish (Florida Museum)

unverified

site-attached, territorial: not schooling fish

S5 the aquarium level and this repository

ours

tank and fish sizes, zForth, our own runs

How to read the chips

verified

the source was opened; the number is on its page

unverified

from a summary, or not opened

ours

our own choice, maths or measurement

Not claimed: that real clownfish school (they do not), that our sweep shows the paper's hysteresis (it is too coarse), or that the Forth was timed inside the engine. Data dated 2026-10-01.